

Name _____ Surname _____ Date ____/____/____

Thermal physics · Mock midterm test · Version C

Time: 50 minutes · Total: 44 marks · Calculator, ruler and sharp pencil allowed. Section IV is in Italian: answer it in Italian or in English.

I · Multiple choice

[8]

Circle one letter for each question. Each question is worth 1 mark.

1. A student writes: "Temperature is the heat that a body has." Which statement describes the error?
 - A Temperature is measured in joules and heat is measured in kelvin.
 - B Heat is a state quantity, so a body can store it.
 - C Temperature is a state quantity, while heat exists only as energy transferred because of a temperature difference.
 - D There is no error: heat and temperature are the same quantity.
2. Two cups of water, X and Y, are each left in contact with the same thermometer until its reading stops changing. The final reading is the same for both cups. Which statement is correct?
 - A X and Y must contain the same amount of heat.
 - B X and Y must have the same mass.
 - C X and Y are at the same temperature only if they are brought into contact.
 - D X and Y are at the same temperature, as the Zeroth Law implies.
3. A furnace operates at 1273 K. Express this temperature in °C.
 - A 1546 °C
 - B 1000 °C
 - C -1000 °C
 - D 1273 °C
4. Which statement about 0 K is correct?
 - A It is the lowest temperature possible: no body can be cooled below it.
 - B It is the temperature at which water freezes.
 - C At 0 K the particles have no energy at all and are perfectly still.
 - D It is only a convention, and temperatures below it exist in deep space.
5. Which change makes a liquid-in-glass thermometer more sensitive?
 - A Making the glass of the bulb thicker.
 - B Calibrating it with more fixed points.
 - C Extending the scale to higher temperatures.
 - D Using a narrower capillary tube, with the same bulb and the same liquid.
6. An aluminium bar 3.0 m long expands by 6.0 mm for a temperature rise of 80 °C. A bar of the same metal, 1.5 m long, is heated by 160 °C. By how much does it expand?
 - A 6.0 mm
 - B 3.0 mm
 - C 12 mm
 - D 24 mm
7. A steel washer at 20 °C just fits over a bolt. The washer is heated to 100 °C while the bolt stays at 20 °C. What happens?
 - A The washer grips the bolt tighter, because the metal expands inwards.
 - B The fit does not change, because only the metal expands.
 - C The washer fits more loosely, because its hole becomes larger.
 - D Nothing can be said without knowing the material of the bolt.
8. A metal block is heated so that each of its three dimensions increases by 0.20 %. Approximately by what percentage does its volume increase?
 - A 0.20 %
 - B 0.60 %
 - C 0.40 %
 - D 0.0080 %

II · Alternative to practical

[14]

A group of students builds a thermometer from a thermistor and an Arduino board. The board gives a reading **N**, a number from 0 to 1023 with no unit (counts). The students calibrate the thermometer in four water baths. In each bath they measure the temperature **T** with a reference liquid-in-glass thermometer with 1 °C divisions, and they take five readings of **N**.

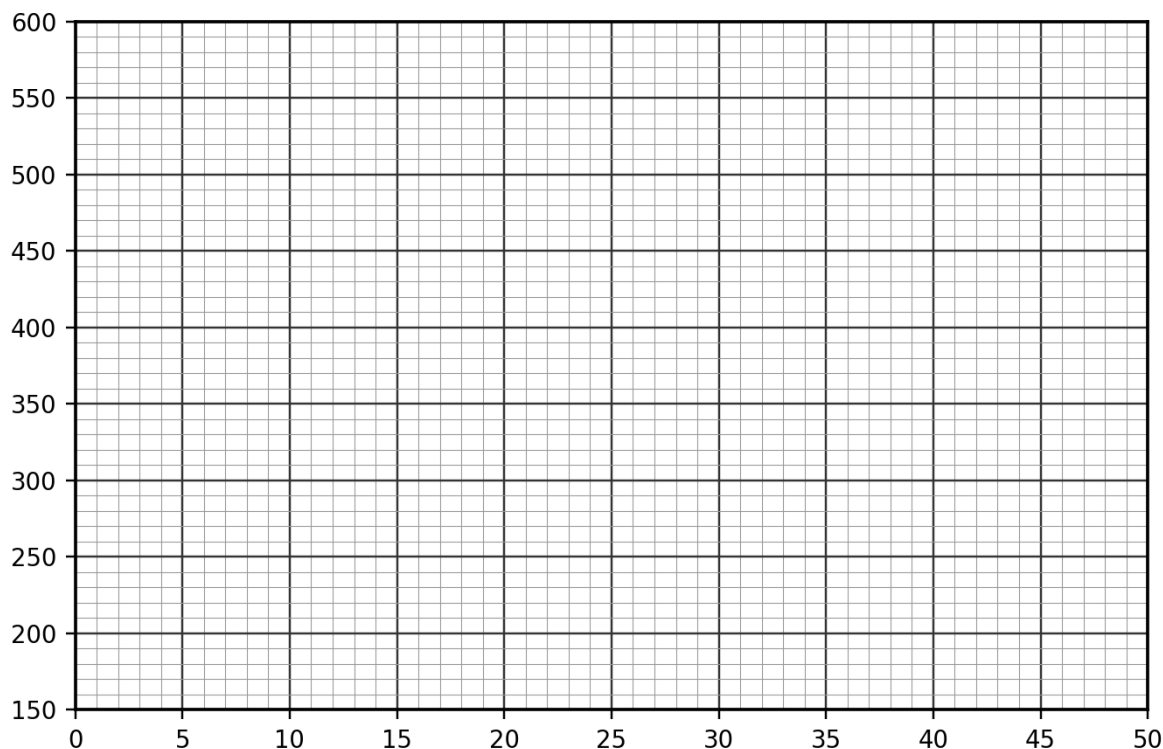
Bath	T / °C	N ₁	N ₂	N ₃	N ₄	N ₅	mean N
1 (ice water)	1.5	238	238	240	242	240	239.6
2 (cold water)	14.0	318	319	318	318	322	319.0
3 (room temperature)	26.0	417	420	417	417	417	417.6
4 (hot water)	43.5	536	533	534	535	494	

(a) State the independent variable and the dependent variable of this experiment. [2]

(b) One of the twenty readings in the table is anomalous. Circle it in the table and give one reason for your choice. [1]

(c) The mean N for one bath has been left blank. Calculate it, ignoring the anomalous reading, and write it in the table. [1]

(d) On the grid, plot the four mean values of N against T. The scales are printed. Mark each point with a small cross. Label both axes with the quantity and the unit. [3]



(e) Draw a single straight line of best fit with a ruler. It should pass as close as possible to all the points, with about as many points above it as below it. Do not join the points. [1]

(f) Use your line to find the gradient m (in counts per $^{\circ}\text{C}$). Show the triangle you use on the graph. [2]

(g) Read from your graph the value N_0 where the line crosses $T = 0^{\circ}\text{C}$. Then write the calibration equation in the form $T = (N - N_0) / m$ with your values. [1]

(h) The students now place the thermometer in a fifth bath, independent of the first four, and read $N = 490$. Use your equation to calculate the temperature of this bath. [1]

(i) The reference thermometer reads 36.8°C in the fifth bath. The client requires the Arduino thermometer to agree with the reference within $\pm 1^{\circ}\text{C}$. Does it meet the requirement? Show your calculation. [1]

(j) Suggest one specific improvement to the students' method that would make the calibration more reliable. [1]

III · Structured questions [14]

1. Temperature scales and heat

(a) Convert (i) 21°C into kelvin and (ii) 250 K into degrees Celsius. [2]

(b) A student says: "The iron bar has a lot of heat, so its temperature is high." Explain what is wrong with this statement. Use the words *temperature*, *heat* and *energy*. [2]

(c) Explain why a temperature of -1 K cannot exist. [1]

2. The thermometer

(a) State the two fixed points used to calibrate a Celsius thermometer, with their values. [2]

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- (b) A thermometer is cooled in a freezer and then dipped into a cup of boiling water. Describe what the liquid column does in the very first moment, and explain it. [2]

3. Thermal expansion

Linear expansion coefficients, α ($\times 10^{-6} \text{ }^\circ\text{C}^{-1}$):

Material	steel	copper	brass	aluminium	glass
α	12	17	19	23	9

- (a) A steel girder is 8.0 m long at $-10 \text{ }^\circ\text{C}$. Calculate its increase in length when it is heated to $40 \text{ }^\circ\text{C}$. [3]

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- (b) A plate of the same steel has an area of 1.20 m^2 at $-10 \text{ }^\circ\text{C}$. Calculate its increase in area when it is heated to $40 \text{ }^\circ\text{C}$. Use $\beta = 2\alpha$ for the area expansion coefficient. [2]
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Anello di rame su un albero d'acciaio. Un'azienda che costruisce motori elettrici deve calettare un anello di rame (massa 0,9 kg, spessore 8 mm) su un albero d'acciaio. A 20 °C il foro dell'anello ha diametro 30,00 mm e l'albero ha diametro 30,03 mm. L'anello viene scaldato in un bagno d'olio con una resistenza da 1,5 kW, mentre l'albero resta a 20 °C; per infilarlo senza forzare serve un gioco di almeno 0,02 mm, cioè il foro deve superare di 0,02 mm il diametro dell'albero. Oltre i 200 °C il rame si riuoce e perde durezza. Si montano 200 anelli al giorno e il motore ruota a 3000 giri/min.

Dati: coefficiente di dilatazione lineare del rame $17 \times 10^{-6} \text{ }^{\circ}\text{C}^{-1}$; dell'acciaio $12 \times 10^{-6} \text{ }^{\circ}\text{C}^{-1}$.

Chiamiamo *interferenza* la differenza (diametro dell'albero – diametro del foro) quando i due pezzi sono alla stessa temperatura: se è positiva, l'albero è più largo del foro e il collegamento è serrato.

(a) Scaldando l'anello, il diametro del foro aumenta, diminuisce o resta uguale? Spiega in una riga. [1]

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(b) A quale temperatura minima va portato l'anello perché scivoli sull'albero con il gioco richiesto? Verifica che il valore sia compatibile con il limite indicato dal costruttore. [3]

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(c) Di quanto aumenta, in percentuale, l'area della sezione del foro alla temperatura trovata in (b)? [2]

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(d) In esercizio l'anello e l'albero arrivano insieme a 150 °C. Calcola l'interferenza a quella temperatura e dì se il collegamento resta serrato, spiegando perché. [2]

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Fine della prova.